



Calculating integers from their prime factor expressions

TASK A: Evaluating products of prime factors

Write the following as integers:

1) $2^3 \times 3^2 \times 5^2 = 1800$

2) $5^3 \times 7 = 875$

3) $2^2 \times 3 \times 11^2 = 1452$

4) $3^3 \times 7 \times 13^2 = 31\,941$

5) $5^2 \times 7^3 \times 19 = 162\,925$

TASK B: Using product of prime factor form

1) Using prime numbers make each of the following products satisfy the condition on the right

- | | | | |
|----|------------------------------|-----------------------------|------------------|
| a) | $3^2 \times 5^3$ | e.g. $\times 2$ | an even number |
| b) | 2^2 | e.g. $\times 3$ | a multiple of 6 |
| c) | $2^4 \times 7$ | e.g. $\times 3^2$ | a multiple of 9 |
| d) | $3 \times 5^3 \times 11^2$ | e.g. $\times 2$ | a multiple of 10 |
| e) | $3^2 \times 5^3 \times 13$ | e.g. $\times 5 \times 3$ | a square number |
| f) | $3^2 \times 5^3 \times 11^4$ | e.g. $\times 3 \times 11^2$ | a cube number |

2) Here is a number written as a product of its prime factors

Do not work it out

$$2^4 \times 5^2 \times 7^2$$

Are the following statements true or false make sure you justify your answers?

- | | |
|----------------------------------|---|
| a) The number is even | <i>True because 2 is a factor</i> |
| b) The number ends in a zero | <i>True, since 2 and 5 are both factors</i> |
| c) The number ends in two zeros | <i>True, the prime factors of 100 are 2^2 and 5^2</i> |
| d) The number has 35 as a factor | <i>True, the prime factor of 35 are 5 and 7</i> |
| e) The number has 50 as a factor | <i>True, the prime factors of 50 are 2 and 5^2</i> |
| f) It is a square number | <i>True, it can be written as $(2^2 \times 5 \times 7)^2$</i> |

3) List all the common factors of:

$$2^4 \times 3 \times 7^2$$

and

$$2^3 \times 3^3 \times 7$$

2	2×3	$2^2 \times 3$	$2^3 \times 3$	3×7
2^2	2×7	$2^2 \times 7$	$2^3 \times 7$	
2^3	$2 \times 3 \times 7$	$2^2 \times 3 \times 7$	$2^3 \times 3 \times 7$	
3				
7				

